

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA S2

B.Tech CSEAI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Architectural Modelling (CO2)

Group No : 8

Team Members :

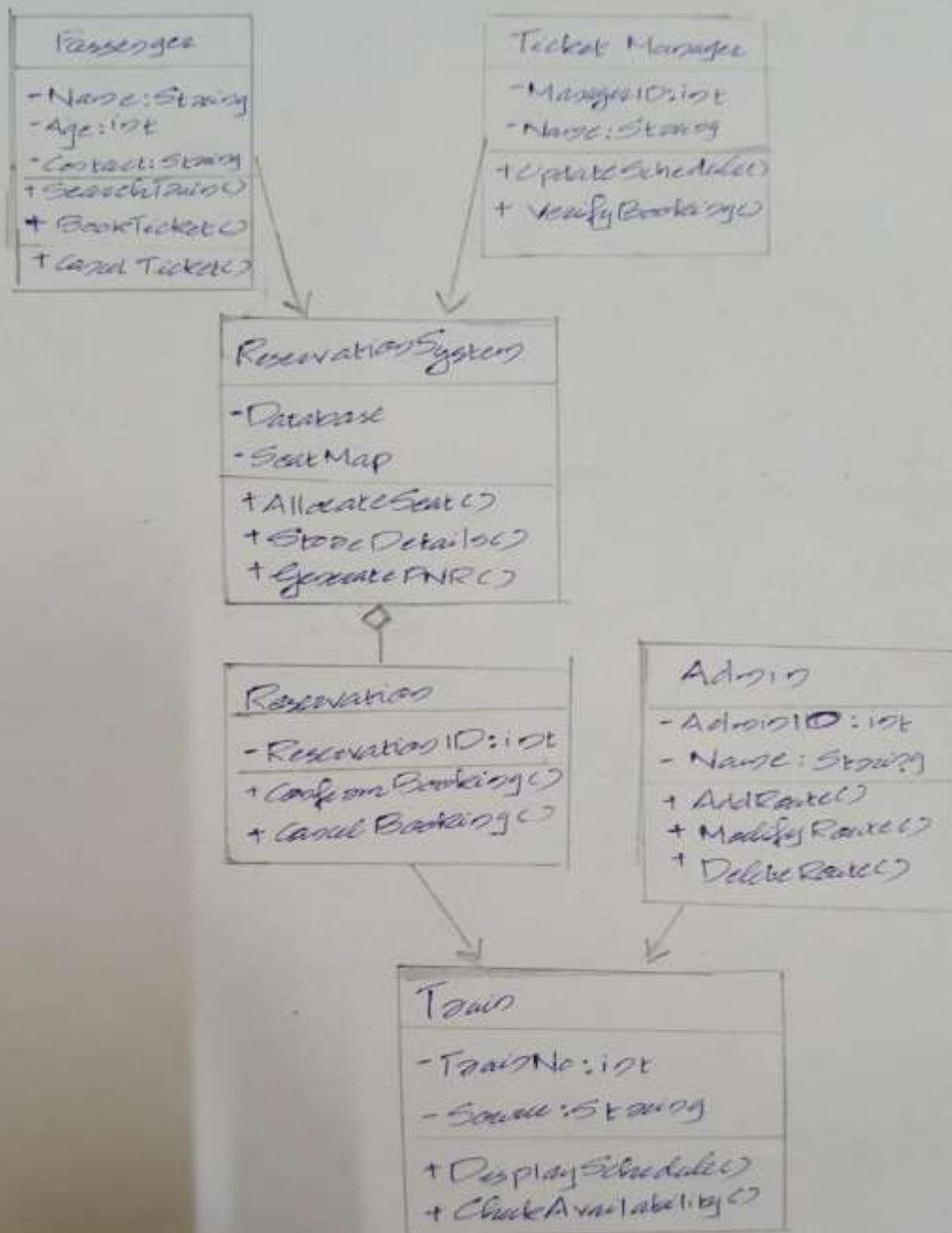
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1. Problem Title : Railway Ticket Reservation System

STRUCTURAL DESIGN

2. Class diagram

Class Diagram



3. Use-case diagram

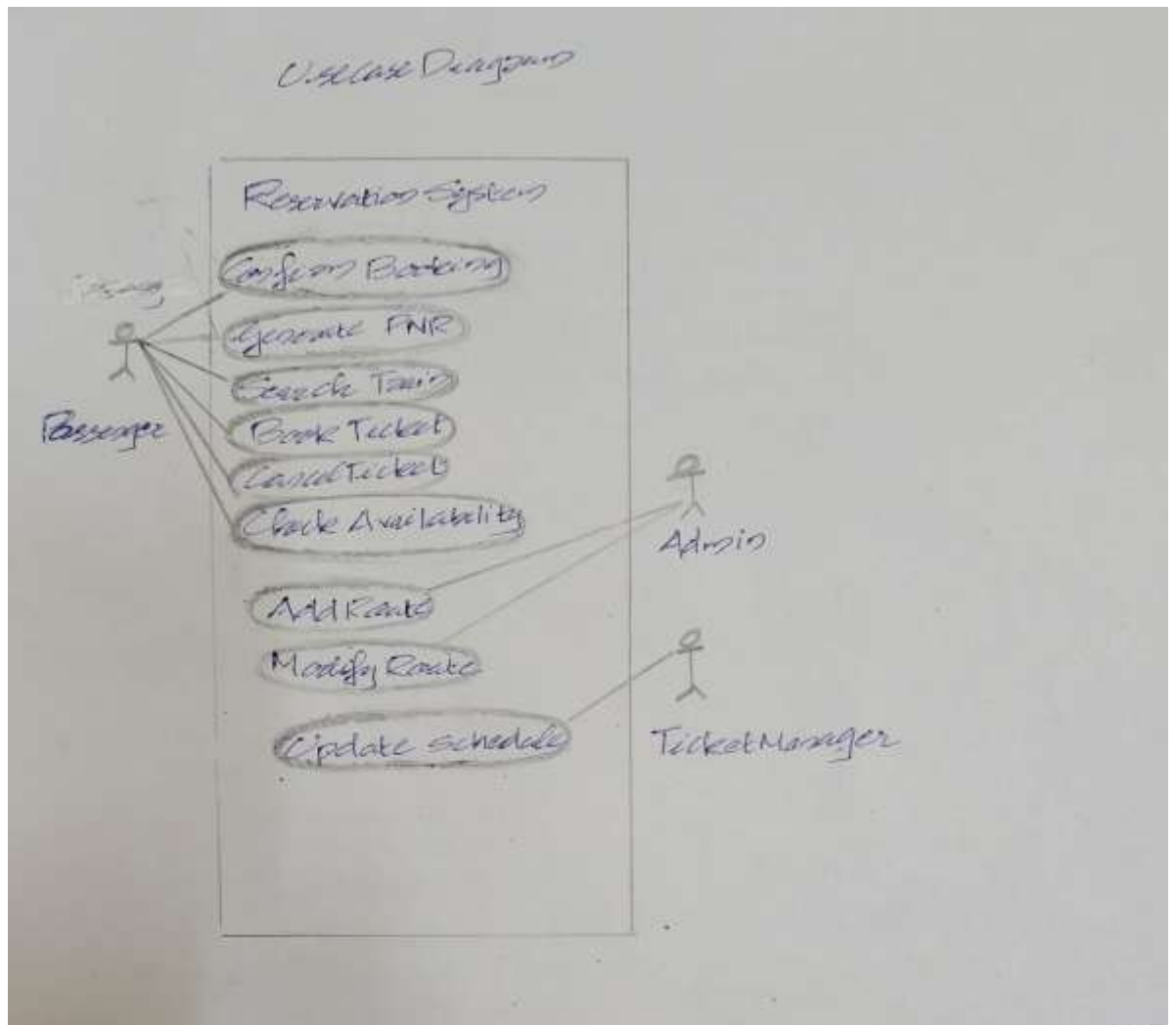
a. List of actors

- 1.Passenger
- 2.Admin
- 3.TicketManager

b. List of use-cases

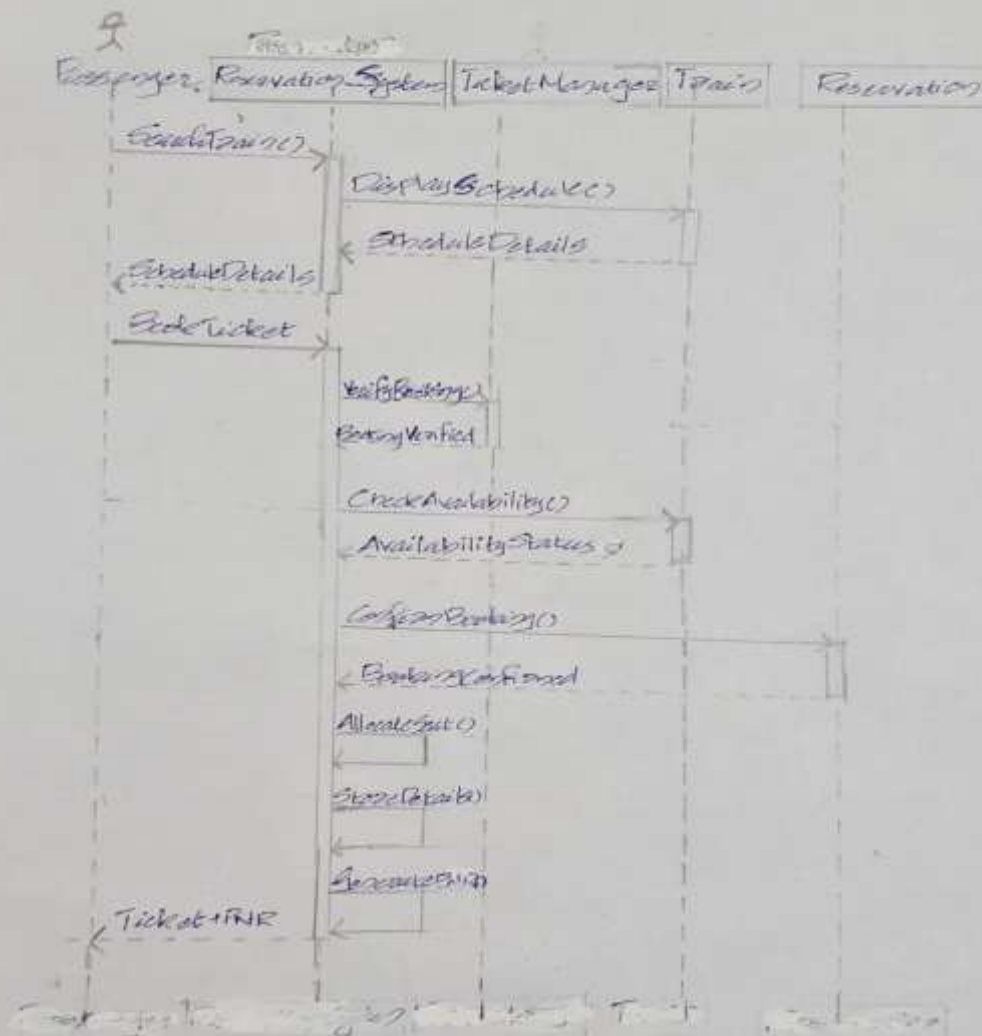
1. Confirm Booking-Passenger receives confirmation of a successful booking.
2. Generate PNR - Passenger obtains a unique Passenger Name Record for the reservation.
3. Search Train- Passenger searches for available trains based on source and destination.
4. Book Ticket- Passenger reserves a seat on a selected train.
5. Cancel Ticket- Passenger cancels an existing reservation using PNR details.
6. Check Availability- Passenger checks whether seats are available on a train.
7. Add Route- Admin adds new train routes into the system. 8. Modify Route- Admin updates existing train route details.
9. Update Schedule- Ticket Manager updates train schedules to keep them current.

c. Use case diagram



4. Sequence diagram (use UML notations)
 - a. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

Sequence Diagram for Booking



A passenger searches for trains to view schedule details and initiates a booking request through the Reservation System. The System uses Ticket Manager to verify the request and check the train for seat availability. Once confirmed, a Reservation is created, the System allocates a seat, stores the details, generates a PNR, and returns the confirmed ticket with PNR to the passenger.

b. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

